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Equal dispersion split-flow mixer

Abstract

Disclosed is a liquid chromatography system and mixer for use therein that includes a first split connected to an inlet, the first split branching the flow of fluid from the inlet into a first path and a second path; a second split connected to an outlet of the first path, the second split branching the first path into a third path and a fourth path; and a third split connected to an outlet of the second path, the third split branching the second path into a fifth path and a sixth path. The first path and the second path are offset by a first predetermined volume, the third path and the fourth path are offset by a second predetermined volume, and the fifth path and the sixth path are also offset by the second predetermined volume.

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Background/Summary

RELATED APPLICATIONS (1) This application is a non-provisional patent application claiming priority to U.S. Provisional Patent Application No. 63/191,081, filed May 20, 2021, entitled “Equal Dispersion Split-Flow Mixer,” which is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The invention relates generally to chromatography. More particularly, the invention relates to an equal dispersion split-flow mixer for use in chromatography systems.

BACKGROUND

(2) Chromatography is a set of techniques for separating a mixture into its constituents. Well-established separation technologies for fluid chromatography systems include HPLC (High Performance Liquid Chromatography), UPLC (Ultra Performance Liquid Chromatography) and SFC (Supercritical Fluid Chromatography). HPLC systems use high pressure, ranging traditionally between 1,000 psi (pounds per square inch) to approximately 6,000 psi, to generate the flow required for liquid chromatography (LC) in packed columns. Compared to HPLC, UPLC systems use columns with smaller particulate matter and higher pressures approaching 20,000 psi to deliver the mobile phase. SFC systems use highly compressible mobile phases, which typically employ carbon dioxide (CO₂) as a principal component.

(3) In many of these fluid chromatography applications, it is desirable to mix fluids that are flowing continuously. For example, in liquid chromatography, a pump is used to deliver precise compositions of solvents to a chromatographic column for the purpose of separating liquid mixtures. The flow composition delivered by the pump can vary in time, and thereby emerge from the pump with compositional disturbances that occur at regular intervals as a result of the pump cycle or other phenomena. These regular intervals can occur at a constant or nearly constant frequency.

(4) In some chromatography applications, additives such as trifluoroacetic (TFA) acid are used in the mobile phase solvents. It is desirable to blend or mix the stream as it flows for the purpose of smoothing out compositional discontinuities that can cause interference with sample detection. Various mixers exist which seek to achieve a desirable mixing of fluids in liquid chromatography systems. For example, large volume mixers exist that mix effectively, but do so with an increase in volume which must be flown through by a fluid or solvent, thereby drastically increasing testing time and diminishing throughput. In contrast, packed-bead LC mixers are inefficient relative to their delay volume, are difficult to manufacture, and are prone to contamination and clogging.

(5) Thus, a mixer that eliminates or reduces the above-described deficiencies would be well received in the art.

SUMMARY

(6) In one exemplary embodiment, a fluid chromatography system comprises: at least one solvent reservoir; a pump connected to the at least one solvent reservoir configured to pump a flow of fluid from the at least one solvent reservoir downstream; a mixer downstream from the pump, the mixer including: an inlet configured to receive the flow of fluid; an outlet configured to provide the flow of fluid downstream from the mixer after the flow of fluid is mixed in the mixer; a first split connected to the inlet, the first split branching the flow of fluid from the inlet into a first path and a second path; a second split connected to an outlet of the first path, the second split branching the first path into a third path and a fourth path; and a third split connected to an outlet of the second path, the third split branching the second path into a fifth path and a sixth path. The first path and the second path are offset by a first predetermined volume. The third path and the fourth path are offset by a second predetermined volume, and the fifth path and the sixth path are also offset by the second predetermined volume. The fluid chromatography system further comprises: a sample injector downstream from the mixer configured to inject a sample into the flow of fluid; a chromatography column downstream from the sample injector configured to perform separation of the sample; and a detector downstream from the chromatography column.

(7) Additionally or alternatively, the mixer further comprises a flow restrictor system located downstream from the third, fourth, fifth and sixth paths, and wherein the flow restrictor system is balanced to provide an equal volume flow rate through each of the third, fourth, fifth and sixth paths.

(8) Additionally or alternatively, the flow restrictor system includes a plurality of coiled restrictors.

(9) Additionally or alternatively, the first predetermined volume is two times the second predetermined volume.

(10) Additionally or alternatively, the mixer further comprises an amplitude controlling feature configured to reduce the amplitude of compositional disturbances.

(11) Additionally or alternatively, the first predetermined volume and the second predetermined volume are each configured to target the same frequency of compositional disturbances in the flow of fluid.

(12) Additionally or alternatively, the first predetermined volume and the second predetermined volume are each configured to target a different frequency of compositional disturbances in the flow of fluid.

(13) Additionally or alternatively, the mixer further comprises a split system that includes the first split, the second split and the third split, and further comprising a recombination system downstream from the split system, wherein the recombination system includes an equal number of recombination points as splits in the split system.

(14) Additionally or alternatively, the difference between the first predetermined volume is at or about 25 percent of the stroke volume of the pump and wherein the second predetermined volume is at or about 12.5 percent of the stroke volume of the pump.

(15) In another exemplary embodiment, a mixer for use in a chromatography system comprises: an inlet configured to receive a flow of fluid; an outlet configured to provide the flow of fluid downstream from the mixer after the flow of fluid is mixed in the mixer; a first split connected to the inlet, the first split branching the flow of fluid from the inlet into a first path and a second path; a second split connected to an outlet of the first path, the second split branching the first path into a third path and a fourth path; and a third split connected to an outlet of the second path, the third split branching the second path into a fifth path and a sixth path, wherein the first path and the second path are offset by a first predetermined volume, and wherein the third path and the fourth path are offset by a second predetermined volume, and wherein the fifth path and the sixth path are also offset by the second predetermined volume.

(16) Additionally or alternatively, the first predetermined volume and the second predetermined volume are each configured to target the same frequency of compositional disturbances in the flow of fluid.

- (17) Additionally or alternatively, the first predetermined volume and the second predetermined volume are each configured to target a different frequency of compositional disturbances in the flow of fluid.
- (18) Additionally or alternatively, the first predetermined volume is two times the second predetermined volume.
- (19) Additionally or alternatively, the mixer includes an amplitude controlling feature configured to reduce the amplitude of compositional disturbances.
- (20) Additionally or alternatively, the mixer includes a flow restrictor system located downstream from the third, fourth, fifth and sixth paths, and wherein the flow restrictor system is balanced to provide an equal volume flow rate through each of the third, fourth, fifth and sixth paths.
- (21) Additionally or alternatively, the flow restrictor system includes a plurality of coiled restrictors.
- (22) Additionally or alternatively, the mixer includes a split system that includes the first split, the second split and the third split, and further comprising a recombination system downstream from the split system, wherein the recombination system includes an equal number of recombination points as splits in the split system.
- (23) In another exemplary embodiment, a method of mixing fluid in a fluid chromatography system comprises: providing a fluid, by at least one fluidic pump, to a mixer; receiving the fluid by an inlet of the mixer; splitting the flow of fluid at a first split into at least a first path and a second path; splitting the flow of fluid in the first path into a third path and a fourth path; splitting the flow of fluid in the second path into a fifth path and a sixth path; offsetting the volume in the first path and the second path by a first predetermined volume; offsetting the volume in the third path and the fourth path by a second predetermined volume; and offsetting the volume in the fifth path and the sixth path by the second predetermined volume.
- (24) Additionally or alternatively, the method further includes targeting the same frequency of compositional disturbances in the flow of fluid with the first predetermined volume and the second predetermined volume.
- (25) Additionally or alternatively, the method further includes targeting a different frequency of compositional disturbances in the flow of fluid with the first predetermined volume and the second predetermined volume.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and further advantages of this invention may be better understood by referring to the following description in conjunction with the accompanying drawings, in which like reference numerals indicate like elements and features in the various figures. For clarity, not every element may be labeled in every figure. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the invention.
- (2) FIG. 1 depicts a block diagram of a liquid chromatography system, in accordance with one embodiment.
- (3) FIG. 2 depicts a schematic view of a mixer for use in the liquid chromatography system of FIG. 1, in accordance with one embodiment.
- (4) FIG. 3 depicts a schematic view of another mixer for use in the liquid chromatography system of FIG. 1, in accordance with one embodiment.
- (5) FIG. 4 depicts a schematic view of another mixer for use in the liquid chromatography system of FIG. 1, in accordance with one embodiment.
- (6) FIG. 5A depicts a graphical representation of a compositional disturbance of a flow of fluid as a function of time through a volume offset split flow mixer prior to a volume offset split, after each

volume offset split before recombination, and after recombination.

(7) FIG. 5B depicts a graphical representation of a compositional disturbance of a flow of fluid as a function of time through a volume offset split flow mixer deploying an amplitude controlling feature prior to a volume offset split, after each volume offset split before recombination, and after recombination.

(8) FIG. 6 depicts a schematic view of another mixer for use in the liquid chromatography system of FIG. 1, in accordance with one embodiment.

(9) FIG. 7 depicts a top view of the mixer of FIG. 6 with channels created in a substrate or surface, in accordance with one embodiment.

DETAILED DESCRIPTION

(10) Reference in the specification to “one embodiment” or “an embodiment” means that a particular, feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment of the teaching. References to a particular embodiment within the specification do not necessarily all refer to the same embodiment.

(11) The present teaching will now be described in more detail with reference to exemplary embodiments thereof as shown in the accompanying drawings. While the present teaching is described in conjunction with various embodiments and examples, it is not intended that the present teaching be limited to such embodiments. On the contrary, the present teaching encompasses various alternatives, modifications and equivalents, as will be appreciated by those of skill in the art. Those of ordinary skill having access to the teaching herein will recognize additional implementations, modifications and embodiments, as well as other fields of use, which are within the scope of the present disclosure as described herein.

(12) In brief overview, the invention relates to a mixer for use in chromatography systems that splits the flow of fluid in order to provide for mixing. Embodiments described herein include one or more of the following desirable traits of a liquid chromatography mixer: ability to operate passively; ease of manufacture, consistent performance, and efficient mixing of a continuous flow stream with respect to pressure drop and delay volume. Mixers described herein are configured to mix longitudinally, i.e. along the flow direction, and may provide for a smaller decay volume than an equivalent packed-bead mixer.

(13) Moreover, mixers described herein may be configured for any setting where a continuous flow of fluid needs to be mixed. Particular embodiments described herein are geared toward fluid chromatography applications, and more particularly to liquid chromatography systems (i.e. HPLC and/or UPLC). However, embodiments may also be applicable to supercritical fluid systems (SFC). Any system in which fluid mixing is required may be applicable to the mixer embodiments described herein.

(14) Mixers consistent with the present invention may further be configured to cancel and/or otherwise reduce fluidic oscillations in composition that depart from a desired composition by one or more frequencies. For example, in cases described herein, one or more pumps (e.g. a single quaternary pump or two binary pumps) located upstream from the mixer may be configured to pump fluid downstream to the mixer. However, prior to mixing by the mixer, the composition expelled by the pump is not sufficiently mixed. Depending on the pump, the composition downstream from the pump oscillate from a desired composition, referred to in the art as “compositional ripple.” Such unwanted compositional variance may occur at regular periodic frequencies dependent on the upstream pump system being used, and may become known to a chromatography system designer such that the mixer may be particularly configured to cancel or reduce one or more the frequencies in composition ripple or variance in accordance with embodiments described herein below.

(15) Mixers described herein may be configured to reduce or cancel this unwanted compositional ripple whether the pump is set to pump a constant amount of solvent, or alternatively set to deliver a gradient. In either situation, there is a desired composition at a given point in time. Any departure

from that desired composition, in the form of a compositional oscillation at a given frequency, is unwanted and may be prevented by the mixers described herein.

(16) Embodiments of the present invention further provide split-flow mixers in which a fluid flow that enters an inlet is split into two or more fluidic paths. One or both of those paths may include a volume offset region configured to delay fluid propagation through the second flow channel relative to the first flow channel. In combination, the paths may be offset by a first predetermined volume. In other words, one of the paths may comprise a volume that is offset (i.e. greater or less than) from another path by a predetermined volumetric amount. In multi-stage designs, the two or more paths may further include additional downstream splits to create additional stage(s). For each additional splitting stage, the two or more paths emerging directly from the split may be offset by a predetermined volumetric amount. The predetermined volumetric amounts within each stage may be the same, but the predetermined volumetric amounts may be different from stage to stage.

(17) The various single or multi stage splits may make up a split system. Connected downstream from the split system may be a flow restrictor system. The flow restrictor system may be balanced in a manner such that the various paths that directly connect to the flow restrictor system include equal volume flow rates when the mixer is deployed. Similarly, downstream from the split system is a recombination system, which may include a number of recombination points that equals the number of upstream split points.

(18) Depending on the particular predetermined volumes to offset the paths after a given split, each stage of the splits may be configured to reduce or cancel these unwanted oscillations in composition at a specific frequency. In combination, any given molecule of fluid introduced to the mixer may flow through one path at each post-split stage before recombining. Overall, the various stages may be configured to significantly cancel most or all of the unwanted compositional ripple or oscillation in the composition of fluid coming from the pump.

(19) FIG. 1 shows an embodiment of an exemplary liquid chromatography system **10** for separating a sample into its constituents. The liquid chromatography system **10** can be an HPLC, UPLC, or the like. The liquid chromatography system **10** includes a solvent delivery system that includes a plurality of solvent reservoirs **18A**, **18B**, **18C**, **18D**. The solvent reservoirs are connected to a gradient proportioning valve **16** which provides the combined solvents to a quaternary pump **12**. The quaternary pump **12** draw solvents through a fluidic conduit, which may be a fluidic conduit, line, tube or channel.

(20) While not shown, other embodiments of the liquid chromatography system **10** contemplated may be a binary pump system having two binary pumps (i.e. using a binary solvent manager BSM system). Thus, the present invention may be included in a BSM system including two high pressure mixing pumps in which frequencies due to the pump cycle cause flow perturbations. In such instances, the frequencies of unwanted compositional fluctuations may be fixed in these BSM systems. Hereinafter, while the quaternary pump **12** is shown, it should be understood that the mixers described herein, and concepts described herein, are applicable to BSM systems as well as quaternary solvent manager (QSM) systems.

(21) The quaternary pump **12** may have a single pair of pump heads and alter the composition via a switching valve upstream of the pump **12**. The quaternary pump **12** may be configured to deliver up to four different solvents (as shown, solvents from reservoirs **18A**, **18B**, **18C**, **18D**) with the switching valve. Compositional ripple as described herein occurs because only one solvent is delivered at a time to the quaternary pump **12** by the gradient proportioning valve **16**. The valve **16** alternates between the solvents rapidly to achieve the commanded composition. However, the solvents may not fully blend in the pump heads. Additionally, during a gradient where the set composition is changing over time, each pump stroke has a different composition. Thus, the quaternary pump **12** in this case creates an undesirable staircase-shaped delivered composition curve that needs additional mixing for proper detection downstream.

(22) Downstream from the quaternary pump **12** may be a mixer, which may be exemplified by any

one or a combination of the mixers **100, 200, 300, 400** described hereinbelow. The mixer **100, 200, 300, 400** may be configured to passively mix the pumped fluid in accordance to the embodiments described herein. While the specific features of mixer **100, 200, 300, 400** is shown in FIGS. 2 and 3 and described herein below, the liquid chromatography system **10** can include any mixer consistent with the mixer embodiments described herein, such as the mixer **200** described in FIGS. 4 and 5, instead of the mixer **100, 200, 300, 400**.

(23) Downstream from the mixer **100, 200, 300, 400** is shown an injector **14**. The injector **14** may be included as a feature of a sample manager or other assembly or sub-system configured to inject a sample into the flow of fluid coming from the mixer **100, 200, 300, 400**. The injector **14** may include an injector valve with a sample loop. The sample manager may control the injection of the sample and may operate in one of two states: a load state and an injection state. In the load state, the position of the injector valve of the injector **14** is such that the solvent manager loads the sample into the sample loop; in the injection state, the position of the injector valve of the injector **14** changes so that solvent manager introduces the sample in the sample loop into the continuously flowing mobile phase arriving from the mixer **100, 200, 300, 400**.

(24) With the injector valve of the injector **14** in the injection state, the mobile phase carries the sample into a column **22**. The chromatography column **22** is in fluidic communication with the injector **14** through, for example, a fluidic tube. The chromatography column **22** may be configured to perform sample separation according to techniques known in the art. Another fluidic tube couples the output port of the column **22** to a detector **24**, for example, a mass spectrometer, a UV detector, or any other detector. Through the fluidic tube, the detector **24** may be configured to receive the separated components of the sample from the column **22** and produce an output from which the identity and quantity of analytes of the sample may be determined. Noise in the absorbance of the separated components over time may be reduced by the mixers described herein.

(25) The liquid chromatography system **10** is shown for exemplary purposes, and the various features shown may be modified, changed or replaced with any features of any known liquid chromatography system without departing from the scope of the invention. For Furthermore, while the invention is shown by way of example with a liquid chromatography system, mixers described herein may be deployed with any fluidic system, including supercritical fluid chromatography (SFC) systems or even non-chromatography applications.

(26) In one exemplary embodiment of the liquid chromatography system **10** shown above, two solvents are delivered from each of solvent reservoirs **18A** and **18B**. The other solvent reservoirs **18C** and **18D** may not be used in this embodiment. The solvent from reservoir **18A** may be water with 0.1% trifluoroacetic acid (TFA). The solvent from reservoir **18B** may be acetonitrile (ACN) with 0.1% TFA. In such an embodiment, more TFA sticks to the column when solvent from reservoir **18A** passes through, less sticks when solvent from reservoir **18B** passes through. In this manner, oscillations in the composition will cause the amount of TFA leaving the column to oscillate. The TFA in the compositions absorbs light at the wavelength the detector **24** is observing. Thus, the mixer **100, 200, 300, 400** is configured to prevent noise waves seen in the baseline of the chromatogram that would otherwise be present if unwanted oscillations in the composition, or “compositional ripple” was present. Such oscillations would interfere with the quantification of the size of sample peaks and thereby are desirable to prevent by the mixer **100, 200, 300, 400** in accordance with embodiments described herein.

(27) FIG. 2 depicts a schematic view of a mixer **100** for use in the liquid chromatography system **10** of FIG. 1, in accordance with one embodiment. The mixer **100** includes an inlet **102** configured to receive a flow of fluid, such as a solvent composition coming from the upstream quaternary pump **12** of FIG. 1. The mixer **100** further includes an outlet **104** configured to provide the flow of fluid downstream from the mixer **100** to the injection location, column **14** and detector **24** shown in FIG. 1, after the flow of fluid is mixed in the mixer **100**. The mixer **100** includes a first split **106** connected to the inlet **102**. The first split **106** may be configured to branch the flow of fluid from

the inlet **102** into a first path **108** and a second path **110**. A second split **112** is shown connected to an outlet of the first path **108**. The second split **112** may be configured to branch the first path **108** into a third path **114** and a fourth path **116**. Likewise, a third split **118** is shown connected to an outlet of the second path **110**. The third split **118** may be configured to branch the second path **110** into a fifth path **120** and a sixth path **122**.

(28) The first path **108** and the second path **110** are shown offset by a first predetermined volume. In the embodiment shown, the first path **108** includes a fluidic volume of 35 microliters (μl) and the second path **110** includes a fluidic volume of 2 μl . Thus, the first predetermined volume offset between the first path **108** and the second path **110** is 33 μl .

(29) Similarly, the third path **114** and the fourth path **116** are shown offset by a second predetermined volume. In the embodiment shown, the third path **114** includes a fluidic volume of 18.5 microliters (μl) and the fourth path **116** includes a fluidic volume of 2 μl . Thus, the second predetermined volume offset between the third path **114** and the fourth path **116** is 16.5 μl .

(30) Finally, the fifth path **120** and the sixth path **122** are shown offset by the same second predetermined volume. In the embodiment shown, the fifth path **120** includes a fluidic volume of 18.5 microliters (μl) and the sixth path **122** includes a fluidic volume of 2 μl . Thus, again the volume offset between the third path **114** and the fourth path **116** is 16.5 μl , the same as the volume offset between the third path **114** and the fourth path **116**.

(31) As shown, the mixer **100** includes two stages of splits, a first stage **124** and a second stage **126**. While the first predetermined volume offset of 33 μl and the second predetermined volume offset of 16.5 μl are also exemplary, it is contemplated that the volume offset at each individual stage of splits is equal. Thus, in the event that the mixer **100** included additional split stages (i.e. a third stage having eight additional paths and/or a fourth stage having sixteen additional paths), it is contemplated that each path pairing would share the same volume offset as the other path pairings in the same stage.

(32) Moreover, the present invention further contemplates two paths extending from each of the splits **106**, **112**, **118**. These split pairings may be particularly beneficial in targeting regular frequencies in compositional disturbances in fluid received by the mixer. However, in other embodiments, each split may break into any number of paths. However, whatever the embodiment, the number of paths extending from a split point may be the same within a given stage. Moreover, the offsets between the two or more paths coming from a split on a given stage may be the same as the offsets between the two or more paths coming from each of the other splits on the same stage.

(33) The respective volumes of the various paths **108**, **110**, **112**, **114**, **116** are exemplary and any volumes are contemplated. However, the volumes shown in the embodiment of FIG. 2 may be particularly useful in targeting specific frequencies of compositional disturbances. For example, at the first stage, the volume offset of 33 μl at the first stage (i.e. the first predetermined volume offset) may target a volume frequency of 132 μl , four times the offset amount. At the second stage, the volume offset of 16.5 μl may also be targeting the same volume frequency of 132 μl . In an embodiment of a liquid chromatography system where the stroke volume of the pump **12** is 132 μl , the volume offset of the mixer may be configured to target the stroke volume. In this embodiment, the difference between the first predetermined volume is at or about 25 percent of the stroke volume of the pump and the second predetermined volume is at or about 12.5 percent of the stroke volume of the pump. In the embodiment shown, the first predetermined volume of the first stage is two times the second predetermined volume at the second stage—this structure (the upstream stage being two times the volume offset as the downstream stage) is configured to target the same frequency of compositional disturbances in the flow of solvent. However, in other embodiments, the first and second predetermined volumes may be different and may be configured to target different frequencies of compositional disturbances in the flow of solvent.

(34) Downstream from the greater volume portions of the third, fourth, fifth and sixth paths **114**, **116**, **120**, **122** is a flow restrictor system **128**. The flow restrictor system **128** may be configured to

restrict flow in a balanced manner in order to provide for ensuring an equal volume flow rate through each of the third, fourth, fifth and sixth paths **114, 116, 120, 122** when the mixer is in use and receiving a flow at the inlet **102**. The flow restrictor system **128** may include four separate flow restrictors **130, 132, 134, 136** located prior to recombining each of the third, fourth, fifth and sixth paths **114, 116, 120, 122**. However, the flow restrictor system **128** may further include flow restrictor sections located downstream from when the third, fourth, fifth and sixth paths **114, 116, 120, 122** begin to converge prior to the outlet **104**. Whatever the configuration, the flow restrictor system **128** may dominate the fluidic resistance of each of the flow paths **114, 116, 120, 122**, thereby restricting volume flow rate in a manner that ensures balanced flow through each of the various flow paths **114, 116, 120, 122**.

(35) The mixer **100** shown can be characterized as including a split system **138** and a recombination system **140**. The split system includes the first split **106**, the second split **112**, the third split **114**. Likewise, the recombination system includes a first and second recombination point **142, 144**, and a third recombination point **146** located downstream from the first and second recombination points **142, 144**. Thus, the recombination system **140** includes an equal number of recombination points **142, 144, 146** as splits **106, 112, 114**.

(36) FIG. 3 depicts a schematic view of another mixer **200** for use in the liquid chromatography system **10** of FIG. 1, in accordance with one embodiment. Like the mixer **100**, the mixer **200** includes an inlet **202** configured to receive a flow of fluid, such as a solvent composition coming from the upstream quaternary pump **12** of FIG. 1. The mixer **200** further includes an outlet **104** configured to provide the flow of fluid downstream from the mixer **200** to the injection location, column **14** and detector **24** shown in FIG. 1, after the flow of fluid is mixed in the mixer **200**.

(37) Unlike the mixer **100**, the mixer **200** includes three successive stages of splits **224, 225, 226**. At the first stage **224**, the inlet **202** is split into one pair of paths (two total paths). At the second stage **225**, the two paths are split into four paths (two pairs of paths). Finally, at the third stage **226**, the four paths are split into eight paths (four pairs of paths). The first two stages include offset volumes that are the same as the first predetermined volume offset of 33 μ l and the second predetermined volume offset of 16.5 μ l found in the mixer **100**. Again, each of these volume offsets (33 μ l at the first stage and 16.5 μ l at the second stage) may be configured to target the frequency embodied by the stroke volume of the pump **12**. However, unlike the mixer **100**, the mixer **200** includes a third stage of four pairs of paths. Each of these pairs of paths includes a third predetermined volume offset of 5 μ l. This 5 μ l volume offset amount, being that it is the offset of the third stage, is configured to target a frequency of 80 μ l. Thus, in this embodiment, the first and second stages may be configured to target one frequency of compositional disturbances in the flow of fluid through the mixer, while the third stage may be configured to target a different frequency of compositional disturbances in the flow of fluid through the mixer.

(38) Additionally, the mixer **200** includes a flow restrictor system **228** that includes eight balanced flow restrictors **230a, 230b, 230c, 230d, 230e, 230f, 230g, 230h** that extend from each of the eight individual paths from the third stage. After the eight paths converge into four outlet paths, the four outlet paths each include a separate coiled restrictor **250, 252, 254, 256**. The coiled restrictors **250, 252, 254, 256** may be configured to help further mix and disperse the fluid. After the four separate outlet paths converge into two outlet paths, the two outlet paths each include a separate coiled restrictor **258, 260**. Finally, after the two outlet paths converge to the outlet **104**, a final coiled restrictor **262** is located in line but immediately upstream from the outlet **204**.

(39) FIG. 4 depicts a schematic view of another mixer **300** for use in the liquid chromatography system **10** of FIG. 1, in accordance with one embodiment. Like the mixers **100, 200**, the mixer **300** includes an inlet **302** configured to receive a flow of fluid, such as a solvent composition coming from the upstream quaternary pump **12** of FIG. 1. The mixer **300** further includes an outlet **304** configured to provide the flow of fluid downstream from the mixer **300** to the injection location, column **14** and detector **24** shown in FIG. 1, after the flow of fluid is mixed in the mixer **300**.

(40) Unlike the mixers **100**, **200**, the mixer **300** splits the flow of fluid into four separate paths. The various paths can have any predetermined volume within greater volume regions **306**, **308**, **310**, **312** in order to target frequencies of compositional disturbances in the flow of fluid through the mixer. In one embodiment, two of the greater volume regions **306**, **312** include a first volume offset configured to target a first frequency, while the two middle greater volume regions **308**, **310** may include volumes to target a second frequency that is different than the first frequency.

(41) Whatever the embodiment, each of the four flow paths of the mixer **300** includes an amplitude controlling feature **314**, **316**, **318**, **320** located downstream from the respective greater volume regions **306**, **308**, **310**, **312**. The amplitude controlling feature is shown as a mini split mixer that includes four additional flow paths. The mini split mixer may also be configured to target a specific frequency of compositional disturbances in the flow of fluid entering the mixer **300**. In other embodiments, the amplitude controlling features may be coiled restrictors, as shown in FIG. 3. Whatever the embodiment, the amplitude controlling features **314**, **316**, **318**, **320** may be configured to reduce the amplitude of the target frequency in a controlled manner so that the resulting amplitudes at the downstream recombination point are equivalent. Moreover, the amplitude controlling features **314**, **316**, **318**, **320** may be configured to disperse fluid in an equivalent manner so that the paths through the mixer include an equivalent or nearly equivalent dispersion.

(42) Both FIGS. 3 and 4 display applying amplitude controlling features **250**, **252**, **524**, **256**, **258**, **260**, **262**, **314**, **316**, **318**, **320** in order to equivalently disperse fluid through each path. The amplitude controlling features, such as those shown in FIGS. 3 and 4, can be applied to any embodiment described herein. For example, amplitude controlling features may be located downstream from the volume offset paths, but upstream from a flow restrictor system.

Alternatively, amplitude controlling features may be located downstream from volume offset paths and also downstream from a flow restrictor system. Finally, an amplitude controlling feature may be located prior to a greater volume path.

(43) FIGS. 5A and 5B display the problem of how an even volumetric split with more dispersion on one side of the volume split might cause a compositional disturbance that is not fully canceled despite going through a frequency targeting volume offset split mixer, such as the embodiments described herein. In particular, FIG. 5A depicts a graphical representation of a compositional disturbance of a flow of fluid as a function of time through a volume offset split flow mixer prior to a volume offset split (stage 1), after each volume offset split before recombination (stage 2), and after recombination (stage 3). The fluid through the mixer inlet is shown at stage 1 to include a composition that is changing at a regular frequency due to a compositional disturbance. At stage 2, the fluid is split into two paths and the phase of the top path is altered relative to the bottom path due to a volume offset. However, the bottom path and the top path include different dispersions resulting in different amplitudes. Thus, after the recombination point at stage 3, the compositional disturbance remains, albeit with a smaller amplitude. Thus, the disturbance has been reduced but not eliminated.

(44) FIG. 5B depicts a graphical representation of a compositional disturbance of a flow of fluid as a function of time through a volume offset split flow mixer deploying an amplitude controlling feature prior to a volume offset split (stage 1), after each volume offset split before recombination (stage 2), and after recombination (stage 3). Like the example of FIG. 5A, the fluid through the mixer inlet in FIG. 5B is shown at stage 1 to include a composition that is changing at a regular frequency due to a compositional disturbance. At stage 2, the fluid is split into two paths and the phase of the top path is altered relative to the bottom path due to a volume offset. Unlike the example in FIG. 5A, the bottom path and the top path include equal dispersions resulting in the same amplitudes. Equivalent dispersions and amplitudes can be created via utilization of an amplitude controlling feature, such as one of the amplitude controlling features **250**, **252**, **524**, **256**, **258**, **260**, **262**, **314**, **316**, **318**, **320**. Thus, after the recombination point at stage 3, the compositional

disturbance is eliminated.

(45) FIG. 6 depicts a schematic view of another mixer **400** for use in the liquid chromatography system **10** of FIG. 1, in accordance with one embodiment. FIG. 7 depicts a top view of the mixer **400** of FIG. 6 with channels created in a substrate or surface, in accordance with one embodiment. While FIG. 6 labels each individual component, FIG. 7 labels only the split and recombination points.

(46) As shown, the mixer **400** includes an inlet **402** configured to receive a flow of fluid, such as a solvent composition coming from the upstream quaternary pump **12** of FIG. 1. The mixer **400** further includes an outlet **404** configured to provide the flow of fluid downstream from the mixer **400** to the injection location, column **14** and detector **24** shown in FIG. 1, after the flow of fluid is mixed in the mixer **400**. The mixer **400** includes a first split **405** connected to the inlet **402**. The first split **405** may be configured to branch the flow of fluid from the inlet **402** into a first path **406** and a second path **407**. A second split **408** is shown connected to an outlet of the first path **406**. The second split **408** may be configured to branch the first path **406** into two additional paths **410**, **411**. Likewise, a third split **409** is shown connected to an outlet of the second path **407**. The third split **409** may be configured to branch the second path **407** into two additional paths **412**, **413**.

(47) The first path **406** and the second path **407** are shown offset by a first predetermined volume. In the embodiment shown, the first path **406** includes a fluidic volume of 135 microliters (μl) and the second path **407** includes a fluidic volume of 2 μl . Thus, the first predetermined volume offset between the first path **406** and the second path **407** is 133 μl . Thus, a first stage **481** of the mixer **400** includes this predetermined volume offset of 133 μl . Similarly, a second stage **482** of the mixer **400** includes a volume offset of 33 μl defined by 35 μl on one path and 2 μl on the other.

(48) Like the mixer **300**, the mixer **400** includes a third stage comprising four split points **414**, **415**, **416**, **417** and eight paths (four pairs) **418**, **419**, **420**, **421**, **422**, **423**, **424**, **425**. The third stage includes a volume offset of 8.25 μl defined by 10.25 μl volume on one path and 2 μl volume on the other. Unlike the mixer **300**, the mixer **400** includes a fourth stage comprising eight split points **426**, **427**, **428**, **429**, **430**, **431**, **432**, **433** and sixteen paths (eight pairs) **434**, **435**, **436**, **437**, **438**, **439**, **440**, **441**, **442**, **443**, **444**, **445**, **446**, **447**, **448**, **449**. The fourth stage includes a volume offset of 4.125 μl defined by 6.125 μl volume on one path and 2 μl volume on the other.

(49) Downstream from the fourth stage includes a flow restrictor system **450** defined by sixteen flow restrictors **451**, **452**, **453**, **454**, **455**, **456**, **457**, **458**, **459**, **460**, **461**, **462**, **463**, **464**, **465**, **466**—one downstream from each of the sixteen paths of the fourth stage.

(50) Moreover the mixer **400** includes a recombination system located downstream from the flow restrictor system **450** that includes recombination points **467**, **468**, **469**, **490**, **470**, **471**, **472**, **473**, **474**, **475**, **476**, **477**, **478**, **479**, **480**. Like the mixers **100**, **200**, **300**, the mixer **400** includes an equivalent split and recombination structure having an equivalent number of split points and recombination points. Finally, the mixer **400** recombines to a single outlet **404**.

(51) The mixer **400** displays that any number of stages are contemplated. Moreover, the mixer **400** further displays that the predetermined volume offset at each stage may be readily configurable to address any particular application.

(52) Methods of mixing fluid in a chromatography system are also contemplated. In particular, methods may include providing a fluid, by at least one fluidic pump, to a mixer, such as one of the mixers **100**, **200**, **300**, **400**. The method may include receiving the fluid by an inlet of the mixer and splitting the flow of fluid at a first split into at least a first path and a second path of the mixer. Methods may include splitting the flow of fluid in the first path into a third path and a fourth path and splitting the flow of fluid in the second path into a fifth path and a sixth path. Methods may include offsetting the volume in the first path and the second path by a first predetermined volume, offsetting the volume in the third path and the fourth path by a second predetermined volume, and offsetting the volume in the fifth path and the sixth path by the second predetermined volume. Additionally, methods may include targeting the same frequency of compositional disturbances in

the flow of fluid with the first predetermined volume and the second predetermined volume. Methods may additionally or alternatively include targeting a different frequency of compositional disturbances in the flow of fluid with the first predetermined volume and the second predetermined volume.

(53) While the invention has been shown and described with reference to specific embodiments, it should be understood by those skilled in the art that various changes in form and detail may be made therein without departing from the spirit and scope of the invention as recited in the accompanying claims. For example, in some embodiments, the removable seal device may not include a support sleeve, or may include a support sleeve that is not made of a metallic material. Other variations are contemplated without departing from the scope of inventions described herein.

Claims

1. A mixer for use in a chromatography system comprising: an inlet configured to receive a flow of fluid; an outlet configured to provide the flow of fluid downstream from the mixer after the flow of fluid is mixed in the mixer; a first split connected to the inlet, the first split branching the flow of fluid from the inlet into a first path and a second path; a second split connected to an outlet of the first path, the second split branching the first path into a third path and a fourth path; and a third split connected to an outlet of the second path, the third split branching the second path into a fifth path and a sixth path, wherein the first path and the second path are offset by a first predetermined volume, wherein the third path and the fourth path are offset by a second predetermined volume, and wherein the fifth path and the sixth path are also offset by the second predetermined volume, and wherein the first predetermined volume and the second predetermined volume are each configured to target the same frequency of compositional disturbances in the flow of fluid.
2. The mixer of claim 1, wherein the first predetermined volume is two times the second predetermined volume.
3. The mixer of claim 1, further comprising a flow restrictor system located downstream from the third, fourth, fifth and sixth paths, and wherein the flow restrictor system is balanced to provide an equal volume flow rate through each of the third, fourth, fifth and sixth paths.
4. The mixer of claim 3, wherein the flow restrictor system includes a plurality of coiled restrictors.
5. The mixer of claim 1, further comprising a split system that includes the first split, the second split and the third split, and further comprising a recombination system downstream from the split system, wherein the recombination system includes an equal number of recombination points as splits in the split system.
6. The mixer of claim 1, wherein the mixer is incorporated into a liquid chromatography system that includes: at least one solvent reservoir; a pump connected to the at least one solvent reservoir configured to pump a flow of fluid from the at least one solvent reservoir downstream; the mixer located downstream from the pump; a sample injector downstream from the mixer configured to inject a sample into the flow of fluid; a chromatography column downstream from the sample injector configured to perform separation of the sample; and a detector downstream from the chromatography column.
7. A method of mixing fluid in a fluid chromatography system comprising: providing a fluid, by at least one fluidic pump, to a mixer; receiving the fluid by an inlet of the mixer; splitting the flow of fluid at a first split into at least a first path and a second path; splitting the flow of fluid in the first path into a third path and a fourth path; splitting the flow of fluid in the second path into a fifth path and a sixth path; offsetting the volume in the first path and the second path by a first predetermined volume; offsetting the volume in the third path and the fourth path by a second predetermined volume; offsetting the volume in the fifth path and the sixth path by the second predetermined volume; and targeting the same frequency of compositional disturbances in the flow of fluid with the first predetermined volume and the second predetermined volume.

8. A mixer for use in a chromatography system comprising: an inlet configured to receive a flow of fluid; an outlet configured to provide the flow of fluid downstream from the mixer after the flow of fluid is mixed in the mixer; a first split connected to the inlet, the first split branching the flow of fluid from the inlet into a first path and a second path; a second split connected to an outlet of the first path, the second split branching the first path into a third path and a fourth path; and a third split connected to an outlet of the second path, the third split branching the second path into a fifth path and a sixth path, wherein the first path and the second path are offset by a first predetermined volume, wherein the third path and the fourth path are offset by a second predetermined volume, and wherein the fifth path and the sixth path are also offset by the second predetermined volume, and wherein the first predetermined volume and the second predetermined volume are each configured to target a different frequency of compositional disturbances in the flow of fluid.

9. The mixer of claim 8, wherein the first predetermined volume is two times the second predetermined volume.

10. The mixer of claim 8, further comprising an amplitude controlling feature configured to reduce the amplitude of compositional disturbances.

11. The mixer of claim 8, further comprising a flow restrictor system located downstream from the third, fourth, fifth and sixth paths, and wherein the flow restrictor system is balanced to provide an equal volume flow rate through each of the third, fourth, fifth and sixth paths.

12. The mixer of claim 11, wherein the flow restrictor system includes a plurality of coiled restrictors.

13. The mixer of claim 8, further comprising a split system that includes the first split, the second split and the third split, and further comprising a recombination system downstream from the split system, wherein the recombination system includes an equal number of recombination points as splits in the split system.

14. The mixer of claim 8, wherein the mixer is incorporated into a liquid chromatography system that includes: at least one solvent reservoir; a pump connected to the at least one solvent reservoir configured to pump a flow of fluid from the at least one solvent reservoir downstream; the mixer located downstream from the pump; a sample injector downstream from the mixer configured to inject a sample into the flow of fluid; a chromatography column downstream from the sample injector configured to perform separation of the sample; and a detector downstream from the chromatography column.

15. A method of mixing fluid in a fluid chromatography system comprising: providing a fluid, by at least one fluidic pump, to a mixer; receiving the fluid by an inlet of the mixer; splitting the flow of fluid at a first split into at least a first path and a second path; splitting the flow of fluid in the first path into a third path and a fourth path; splitting the flow of fluid in the second path into a fifth path and a sixth path; offsetting the volume in the first path and the second path by a first predetermined volume; offsetting the volume in the third path and the fourth path by a second predetermined volume; offsetting the volume in the fifth path and the sixth path by the second predetermined volume; and targeting a different frequency of compositional disturbances in the flow of fluid with the first predetermined volume and the second predetermined volume.
